Mathematical Modelling — Tutorial 1

How these tutorials work. Each tutorial has pen-and-paper parts and a *computational task* that you carry out in whatever language you prefer (MATLAB is recommended; Python or Julia are equally fine). The point of the code is not the language, but that a classmate can regenerate your figure from your folder. Each week, you add a folder `weekNN/<username>/` to the course tutorials repository on GitHub, containing a half-page `report.md`, the figure(s) requested, the code that produced them, and a short `README` giving the command to regenerate them. You submit this as a *pull request* and review one classmate's in return (see the repository's `CONTRIBUTING.md`). Parts marked *Challenge* are optional, for the motivated or when time allows. Chapter and appendix references are to the course notes. Every `report.md` should also close with one sentence of *interpretation*: what does your result mean for the real system you modelled, and not merely as a piece of mathematics?

Before you start — GitHub setup. The computational work is submitted through GitHub. *This week, before the tutorial:* (1) create a free GitHub account if you do not already have one, at `github.com/signup`; (2) email me your GitHub username so I can add you to the course organisation — it is *private*, so you must be invited and cannot join on your own; (3) accept the invitation that then arrives by email. Full step-by-step instructions — creating the account, joining the organisation, installing Git, and a first-time guide for those new to it — are in the **"Start here"** document in the **Modules** section on Canvas. *Read it first if you have never used GitHub.* You do not need to arrive knowing Git: in the tutorial, we will run a short, hands-on introduction to the few commands you need (clone, commit, push, and opening a pull request), and by the end, you will have made your first submission. All you need beforehand is an account and an accepted invitation.

Tutorial 1 (Week 1 — Dimensional analysis in practice)

Notes: Chapter on dimensional analysis. This week also serves as your onboarding to the course repository.

1. Rederive, by dimensional analysis alone, the period of a simple pendulum $T = \sqrt{L/g}\,\Phi(\theta_0)$, stating clearly what the method fixes and what it leaves in the unknown function Φ.
2. Pick one further system — the period of a mass on a spring, the speed of deep-water waves, or Kepler's third law — and carry out the same analysis: list the variables, form the dimensionless groups, and state precisely what is predicted and what is left undetermined.
3. A rigid-body (compound) pendulum — an extended rigid object swinging about a pivot — has small-amplitude period $\mathcal{T} = 2\pi\sqrt{L_{\text{eff}}/g}$, where g is gravity and $L_{\text{eff}} = I/(mR)$ is its *effective length*, built from the body's mass m, its moment of inertia I about the pivot, and the pivot-to-centre-of-mass distance R. Without computing any moment of inertia, explain why dimensional analysis already guarantees the $\sqrt{L_{\text{eff}}/g}$ dependence. What are the dimensions of I?

Computational task. Taylor's blast-wave argument predicts $R = C\,(Et^2/\rho)^{1/5}$ for the radius R of a blast at time t. Generate a synthetic "photographic" data set: fix $\rho = 1.2$ and take the constant $C = 1$ (its theoretical value), choose an energy E, compute $R(t)$ at a dozen times, and add some random noise to mimic measurement error. Now *pretend you do not know* E: fit a straight line to $\log R$ versus $\log t$, check that the slope is close to $2/5$, and recover E from the intercept. Report how close your estimate is to the true E. Which estimation is more accurate, the slope or E? Is there any unexpected result in the estimates? If yes, can you explain why?

Repo deliverable.

1. Your fitting code;
2. a log–log plot showing the data and the fitted line;
3. a `README.md` giving the command(s) to regenerate the figures, and
4. a `report.md` stating the recovered slope and energy — closing with what this shows, conclusions, etc.

This is your first pull request—get the workflow working now while the mathematics is easy.